

JAWAD HAMZA

I build the systems people
run their day on.



Blood banks, hospital wards, restaurant counters, retail tills, genetics labs.
Not demos — working software deployed into live operations, where a wrong stock
figure loses money and a mislabelled blood bag is a clinical incident.

REGISTER A	Systems in production
REGISTER B	Platforms & applied AI
REGISTER C	Web, product & brand
REGISTER D	Media & video production
REGISTER E	Foundations & learning lab
REGISTER F	Record

PROFILE

Software developer, forward deployed

I am a software developer based in **Mirpur, Azad Kashmir**, with two years of freelance delivery and a year running operations and project management inside a software house. Most of my work is **line-of-business software for organisations that cannot afford it to be wrong** — a blood centre, a district hospital, a university genetics lab, a shop floor.

That context shapes how I build. I work close to the people who will use the system, which is why my current role at the Regional Blood Centre is a **forward deployed** one: the requirements live in the departmental manuals and in what the technician actually does at the bench, not in a specification document. It also means I care about the unglamorous parts — audit trails, expiry alerts, double-entry that reconciles, backups that restore.

My primary stack is **Python and Django**, extended by ASP.NET Core and C# for the .NET estates I have worked in, and Next.js and TypeScript where the front end carries real weight. I work fluently with **generative and agentic AI tooling** — Claude Code and OpenAI's Codex — as part of how I ship, and I build self-hosted AI infrastructure so institutions keep their own data. Before software I trained as a video editor and designer, which is still how I think about interfaces.



SYSTEMS BUILT	LIVE DEPLOYMENTS	PRIMARY STACKS	ALSO DELIVERS
20+ Operational platforms, web products, AI tooling	4 Blood centre, hospital, genetics lab, retail floor	3 Django · ASP.NET Core Next.js / TypeScript	2 Brand identity systems and video production

REGISTER A

Systems in production

Software that other people depend on to do their job. Each entry lists where it runs and which modules actually shipped. Screenshots are captured from the running application, not from mockups.

IN PRODUCTION

RBC · MIRPUR AJK

Blood Bank Management System

End-to-end traceability for a regional blood centre: from donor registration and mobilisation drives, through collection, serological screening and component processing, to cross-matched issue at the ward. Every bag carries an audit trail from the arm it came from to the patient it reached.

DEPLOYED

Regional Blood Centre (RBC), Mirpur, Azad Kashmir

ROLE

Forward deployed engineer & developer

BUILT TO

The centre's departmental manuals — mobilisation, collection, medical officer, processing, serology, distribution, and the blood bank KPI set

ASP.NET Core 8 MVC · C# · Entity Framework Core · SQL Server · Razor

SHIPPED MODULES			11 CONTROLLERS
Donors REGISTRY & ELIGIBILITY	Donations COLLECTION EVENTS	Blood Bags INVENTORY & EXPIRY	Blood Test Reports SEROLOGY SCREENING
Transfusions ISSUE & CROSS-MATCH	Patients RECIPIENT RECORDS	Administration USERS & ROLES	Audit Logs CHAIN OF CUSTODY
Modules FEATURE CONTROL	Account AUTHENTICATION	Home OPERATIONS DASHBOARD	

IN PRODUCTION

QAU · GENETICS

Neurogenetics Research LIMS

A laboratory information management system for translational genetics. Families are modelled as real pedigrees — individuals, relationships, probands — so a sample is never just a barcode but a person inside an inheritance pattern. Tracks samples through extraction and sequencing, records variants against a disorder dictionary, and exports family data for analysis.

BUILT FOR

Genetics labs, Quaid-e-Azam University · originally commissioned by Prof. Sundas Farooq

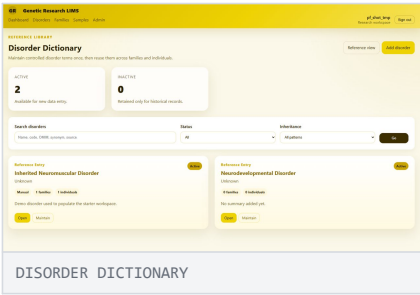
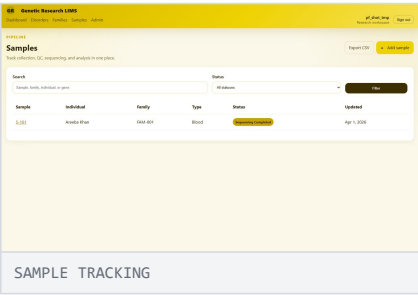
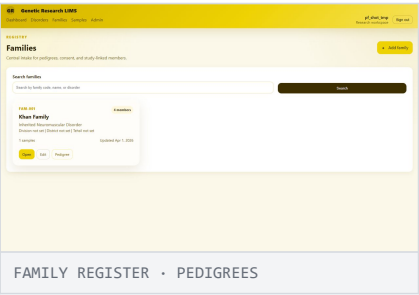
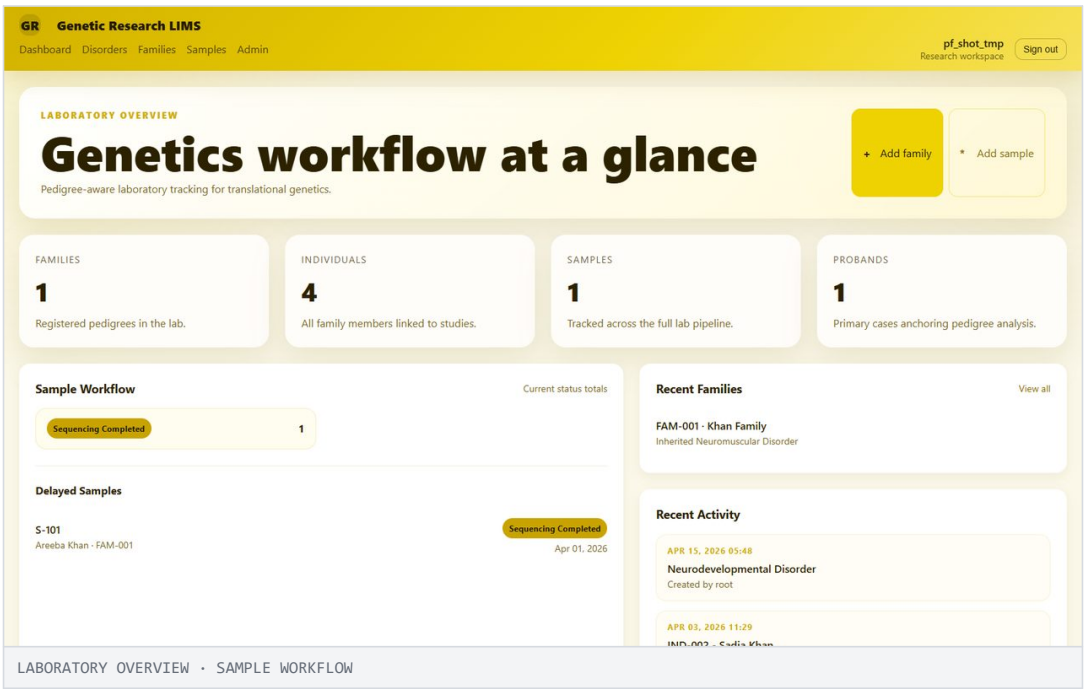
CORE

Pedigree graph, sample workflow states, variant calls, stored file vault, disorder reference

NOTABLE

Interactive pedigree renderer driven from live relationship data

Django · Python · MySQL · Class-based views · Custom auth model



IN PRODUCTION

DHQ · MIRPUR AJK

DHQ Hospital Management System

Contributed to the hospital-wide information system for the District Headquarters Hospital — a multi-area ASP.NET application covering the clinical, diagnostic, pharmacy and finance sides of the hospital under one roof.

DEPLOYED

DHQ Hospital, Mirpur, Azad Kashmir

ROLE

Contributing developer — backend and data layer

SCALE

18 functional areas across clinical, diagnostic and financial operations

ASP.NET MVC · C# · MS SQL Server · Areas architecture

FUNCTIONAL AREAS		18 MODULES	
Dashboard	Appointments	Doctor	Patient History
Vital	Specimen	Labs	Medicine
Pharmacy	Accounts	Finance	Individual Package
Working Sessions	Performance	Building	Country
Province & City	Configuration		

PILOT / SAAS

VENDORA

Vendora — POS, Inventory & Accounts

A retail platform that refuses to separate the till from the books. Barcode checkout, purchase orders, expenses and party ledgers all post into one double-entry accounting core, so the trial balance and profit & loss are generated from actual sales rather than re-keyed at month end. GST is applied automatically from the product category matrix and locked from settings so cashiers cannot drift the tax.

BUILT FOR

Grocery, pharmacy, electronics and garment retailers

ACCOUNTING

Trial balance · profit & loss · balance sheet · account ledgers · party statements

OPERATIONS

POS checkout, stock control, purchases, expenses, multi-user roles, backups

Django · Python · SQLite / PostgreSQL · Thermal receipts · Barcode scanning

POS CHECKOUT • SCAN, CART, GST

OPERATIONS DASHBOARD

TRIAL BALANCE • DOUBLE ENTRY

PROFIT & LOSS STATEMENT

PRODUCT CATALOGUE & STOCKREPORTS CENTRE

PURCHASES & SUPPLIERS

DELIVERED RMS

Restaurant Management System

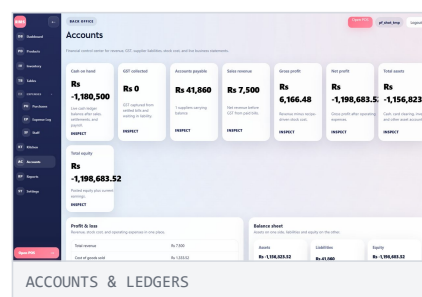
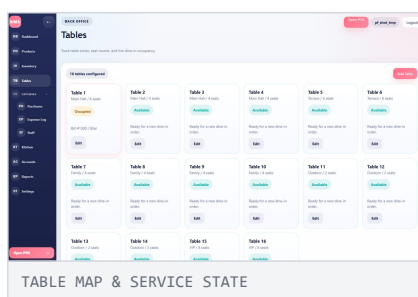
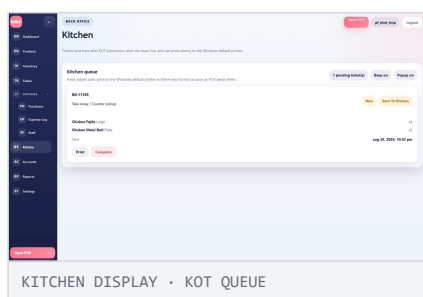
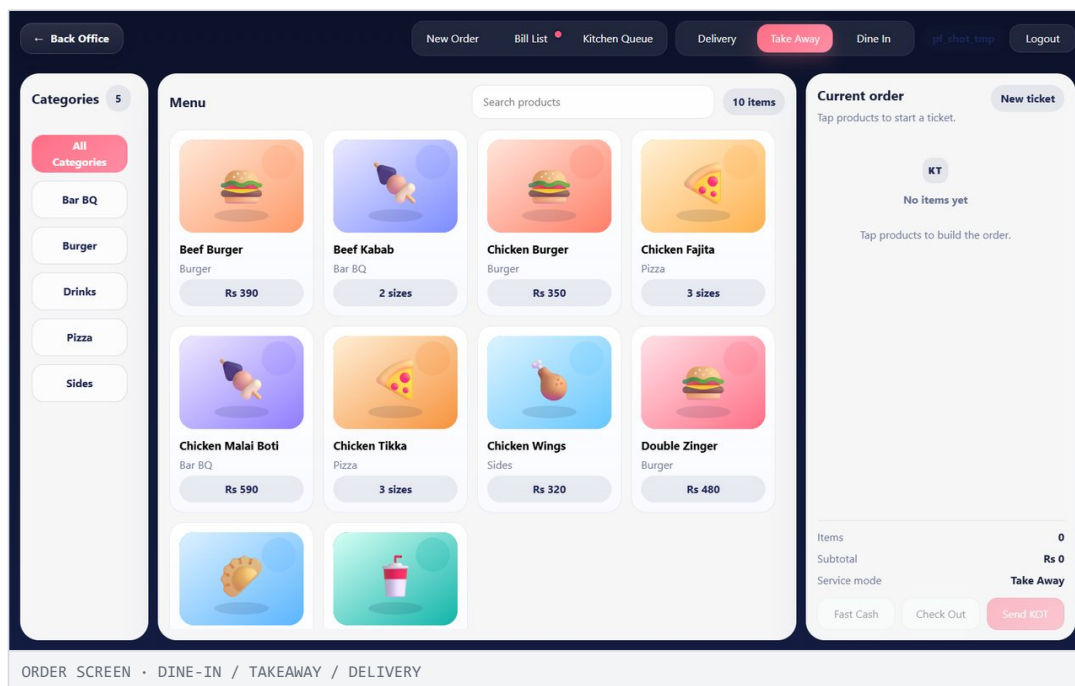
Front of house and back office in one system. A touch-first order screen handles dine-in, takeaway and delivery, fires KOTs straight to a kitchen display, and settles against table state — while the back office keeps recipes, inventory depletion, staff shifts, purchases and the accounts ledger in step behind it.

FRONT OF HOUSE Order screen, bill list, kitchen queue, table map, fast cash

BACK OFFICE Menu & categories, inventory, staff, purchases, expenses, accounts, reports

INTERFACE Built for touchscreen terminals and a separate kitchen display

Django · Django REST Framework · Python · SQLite · KDS



REGISTER B

Platforms & applied AI

Self-hosted, provider-neutral AI infrastructure and agent systems — built so an institution keeps its own data on its own hardware. Retrieval pipelines, multi-agent orchestration, and speech systems that run against local models.

IN DEVELOPMENT

STORE LISTEN

Store Listen — Retail Conversation Intelligence

Records in-store customer conversations, transcribes and translates them live across English, Spanish, French and Urdu, and turns them into coaching for the shop owner. A slang vocabulary teaches the system how customers actually talk; compliance rules attach a dollar value to every missed action. All text is PII-redacted before it reaches a model.

- ENGINES
- Browser Web Speech API for instant words, or chunked cloud STT for any browser
- SECURITY
- Provider keys AES-256-GCM encrypted at rest, never returned to the browser
- OUTPUT
- Live coaching, owner dashboard, daily report, custom metrics and flags

Next.js 16 · React 19 · TypeScript · Zod contracts · SQLite · Gemini / OpenAI / Ollama

WORKSPACE BOUNDARIES		MONOREPO
apps/web RECORDER, DASHBOARD, REPORTS	apps/api AUTH, PROVIDERS, PERSISTENCE	packages/contracts VERSIONED ZOD SCHEMAS
Vocabulary SLANG & MIS-HEARING MAP	Compliance rules TRIGGER → EXPECTED ACTION	Analytics rules GUIDELINES, METRICS, FLAGS

IN DEVELOPMENT

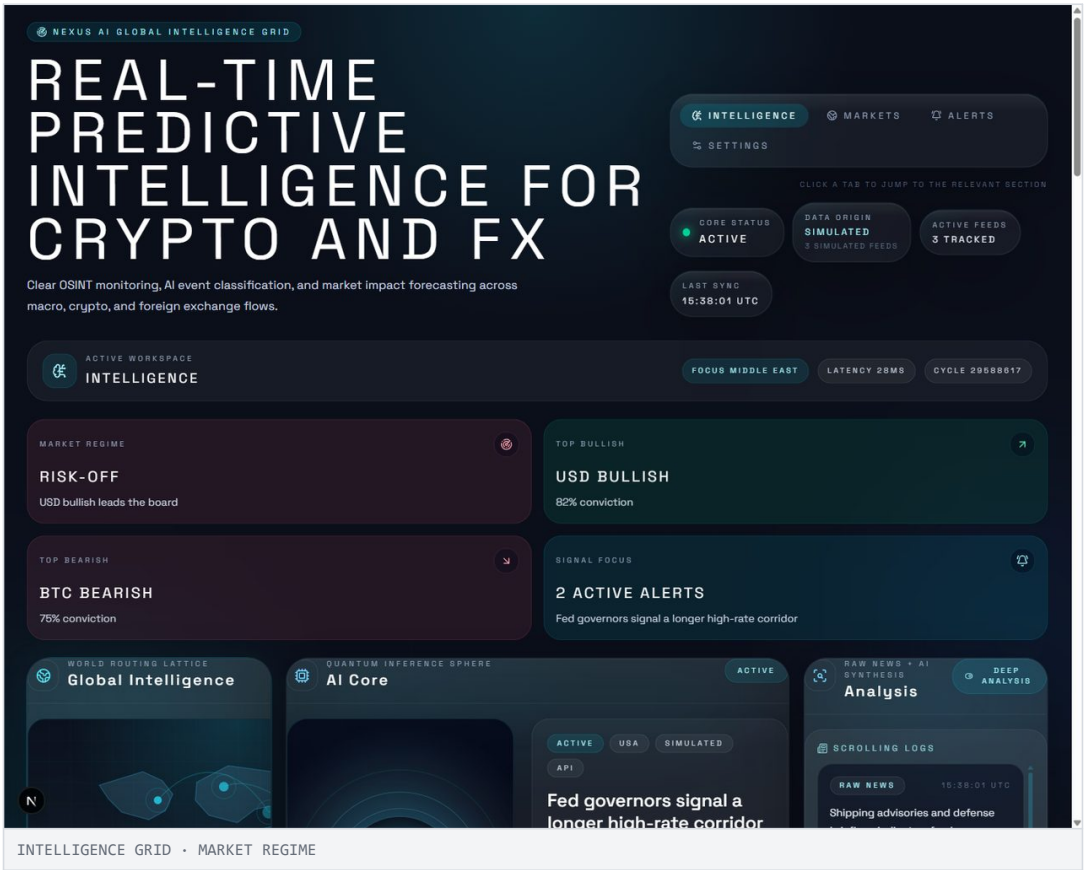
NEXUS AI

Nexus AI — Market Impact Intelligence

An OSINT monitoring grid that ingests open-source event feeds, classifies them, and forecasts market impact across macro, crypto and foreign exchange. Built as a Django event engine behind a Next.js intelligence dashboard with live regime read-outs and alerting.

- PIPELINE
- Feed ingest ☐ event engine ☐ classifier ☐ analytics ☐ impact forecast
- SURFACE
- Intelligence, Markets, Alerts and Settings workspaces

Django · Next.js · Recharts · Tailwind · Python classifiers



IN DEVELOPMENT

UNIVERSITY AI

University AI

Self-hostable, provider-neutral AI platform for university communities. A modular monolith — API, web portal and worker deploy independently while Postgres, Redis and inference stay replaceable. Persisted RBAC, streaming chat through an Ollama gateway, audit records and a containerised local stack.

FastAPI · PostgreSQL · Redis · Docker Compose · Ollama · RBAC

IN DEVELOPMENT

LOCAL AI PLATFORM

Private RAG Platform

A production-oriented, self-hosted ChatGPT alternative for private infrastructure. Admin-managed knowledge bases with document chunking and pgvector embedding, similarity search with source citations, per-user file uploads, project workspaces and an audit dashboard.

FastAPI · pgvector · Celery · SSE streaming · Next.js 14 · Alembic

IN DEVELOPMENT

IDEAFLOW

IdeaFlow

A living project-memory platform. Capture unstructured thoughts, let an LLM extract intent and entities, connect them to existing project knowledge, and detect conflicts before a decision is silently overwritten — then generate architecture, flowcharts, roadmaps and execution-ready prompts traceable back to source.

Next.js App Router · FastAPI · Supabase · pgvector · RLS policies

DELIVERED

AURA

AURA — Desktop AI Operating System

A Jarvis-inspired multi-agent desktop assistant. A supervisor agent routes every request; knowledge and execution agents parse strict JSON intent through local Ollama models and launch real tools for files, browser and applications, with durable memory across sessions.

Python · PyQt6 · Ollama · Multi-agent · Qwen 2.5

DELIVERED

JARVIS VOICE

Jarvis Voice Control

A Windows voice assistant with a deliberately closed command set — it will not run arbitrary shell commands. Speech recognition runs in a local browser page; a Node server drives the PC; a local Qwen model decides whether to run a safe command, open a search, ask for clarification, or simply answer.

Node.js · Web Speech API · Ollama · PowerShell

DELIVERED

FLOW FACTORY

Codex n8n Flow Factory

A workshop that lets a coding agent author n8n workflows as versioned JSON, validate them against known-good templates, map credentials, and import them into a self-hosted n8n workspace as inactive drafts — so the last human step is pressing Execute.

Node.js · n8n public API · JSON schema validation · PowerShell GUI

DELIVERED

TOOLVERSE

ToolVerse

60+ free utility tools for documents, business, images, data, colour and text — every one running entirely in the browser. No uploads, no signup, no tracking: files never leave the device. Installable as a PWA.

Client-side only · PWA · PDF & DOCX processing · MIT

DELIVERED WAKEFORCE

WakeForce — Android

A mission-based alarm app for Android. Alarms are dismissed by completing missions rather than tapping snooze, with progress rings, streak statistics and an emergency override path. Built natively in Kotlin with Jetpack Compose and instrumented UI tests.

Kotlin · Jetpack Compose · MVVM · Gradle KTS · Compose UI tests

REGISTER C

Web, product & brand

Marketing sites, WordPress themes, product interfaces and full identity systems — from logo grid and colour spec through to the shipped, responsive build.

LIVE FUTURESPEACE

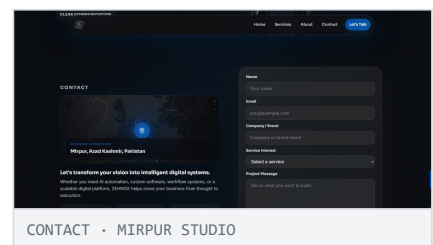
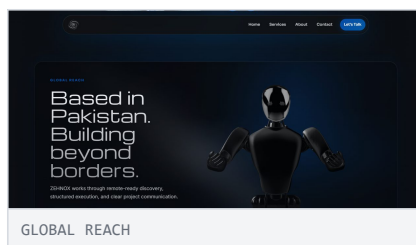
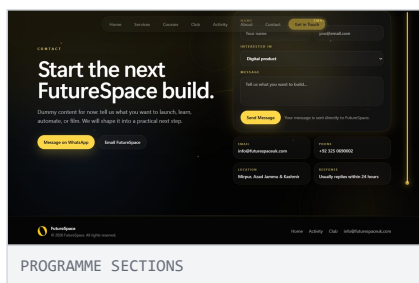
FutureSpace

The digital home of the software house where I ran delivery — a React/Vite site with a particle-driven hero, a courses and club programme, an activity feed and a global-reach section, shipped alongside a matching WordPress theme. Rebuilt and QA'd down to 360 px across every page.

SCOPE Site architecture, front-end build, WordPress theme, responsive QA

SECTIONS Home, Services, Courses, Club, Activity, About, Team, Contact

React · Vite · TypeScript · WordPress theme · Vercel



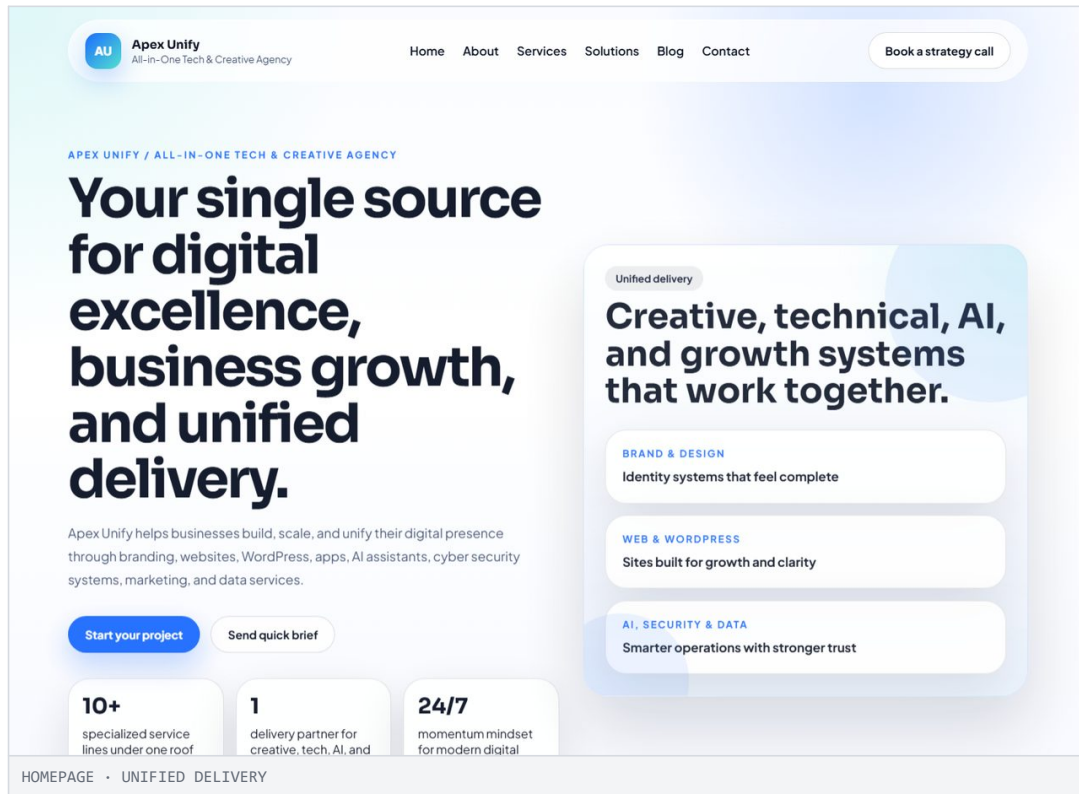
DELIVERED APEX UNIFY

Apex Unify

A full brand and web system for an all-in-one tech and creative agency: identity, animated logo, marketing site and a customised WordPress theme covering ten service lines from brand design through AI assistants and cyber security.

DELIVERED Identity, animated mark, site design and build, WordPress theme package

HTML / CSS / JS • WordPress • Brand identity • Motion



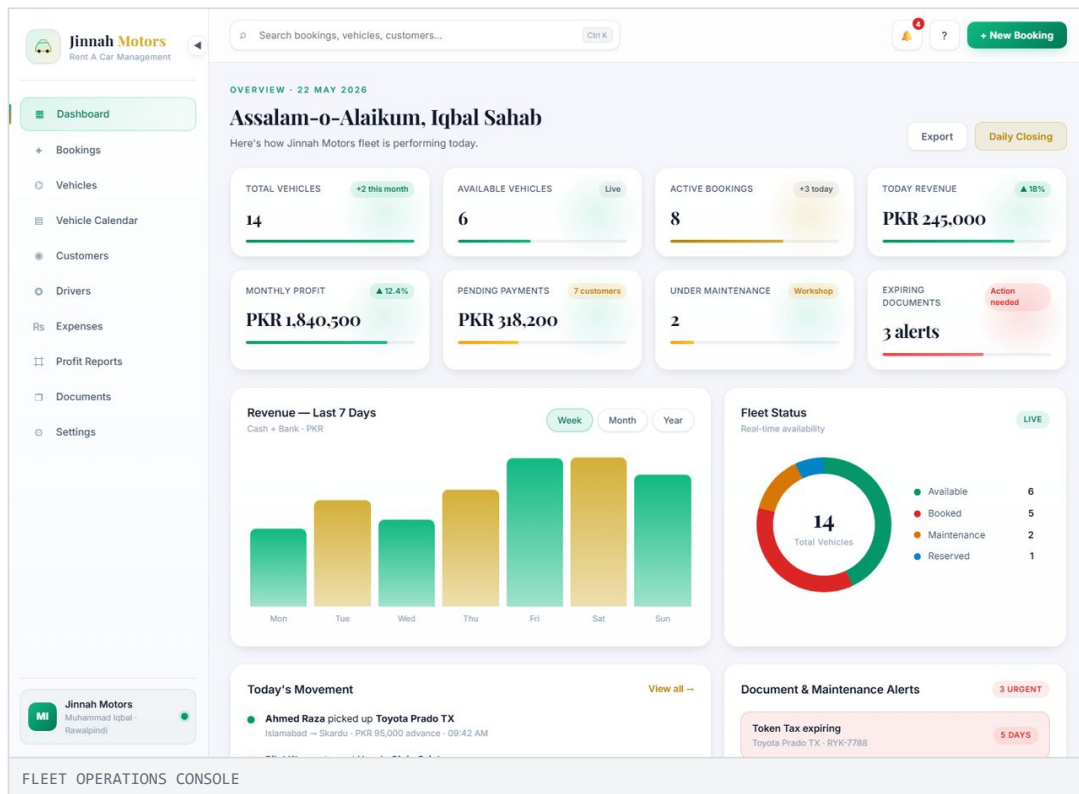
CONCEPT BUILD JINNAH MOTORS

Rent-A-Car Management

A fleet operations console for a Rawalpindi rent-a-car business: bookings, vehicle calendar, drivers, expenses, profit reports, and — the part that matters locally — token tax and document expiry alerts before a vehicle is stopped at a checkpoint.

FOCUS Fleet availability, daily closing, PKR profit reporting, document alerts

HTML / CSS / JS • Dashboard UX • PKR reporting



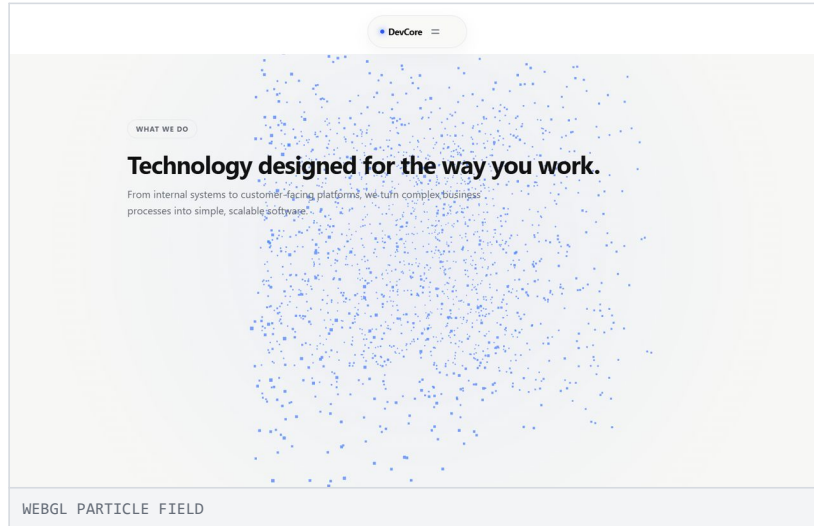
DELIVERED

DEVCORE

DevCore

Agency site for Star Solutions built on a WebGL particle field with GSAP scroll choreography — three.js background, scroll-triggered reveals, and a restrained type system holding it all together.

Three.js · GSAP ScrollTrigger · WebGL



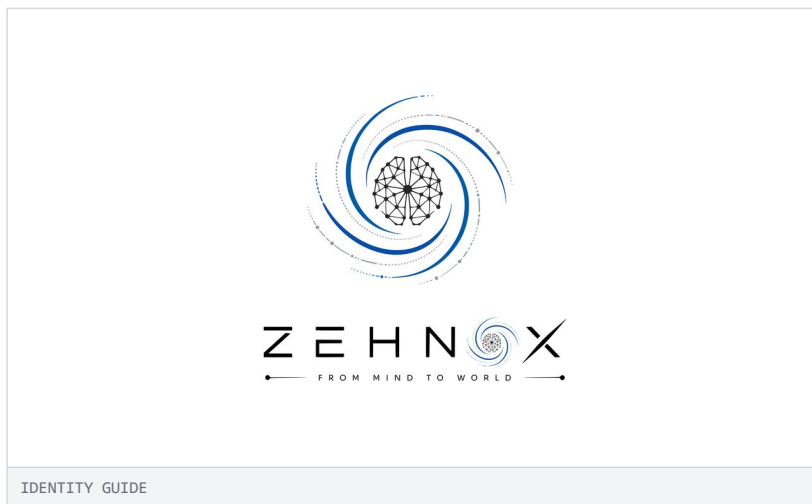
DELIVERED

ZEHNOX

ZEHNOX Identity System

A complete brand identity — primary mark, logo variants, colour and typography specification, a ten-page identity guide, motion treatment and a documented website structure under the line “From Mind to World”.

Adobe Illustrator • Photoshop • Identity guide • Motion



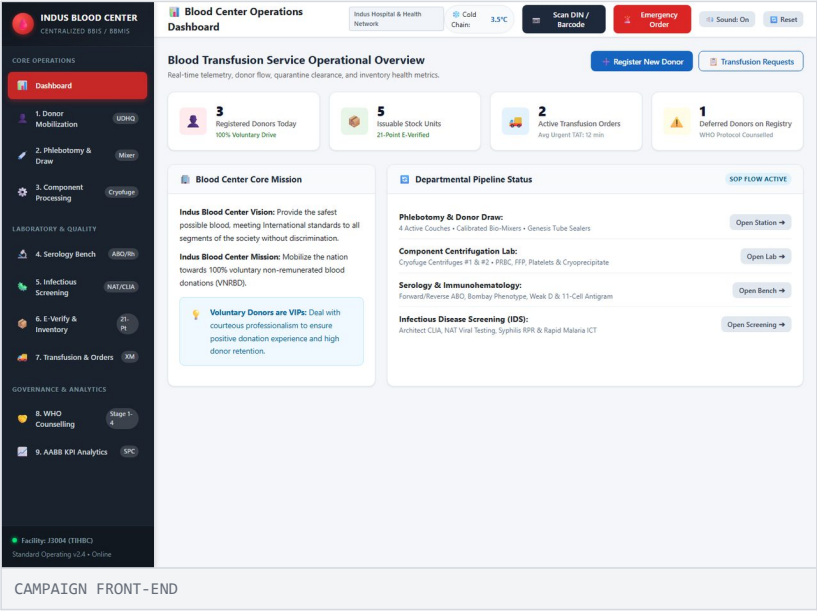
DELIVERED

BLOOD DRIVE

Blood Donation Campaign Sites

Two public-facing campaign front-ends for blood donation drives — donor sign-up, eligibility guidance and drive information, designed to be readable on a phone at a camp table.

HTML / CSS / JS • Responsive

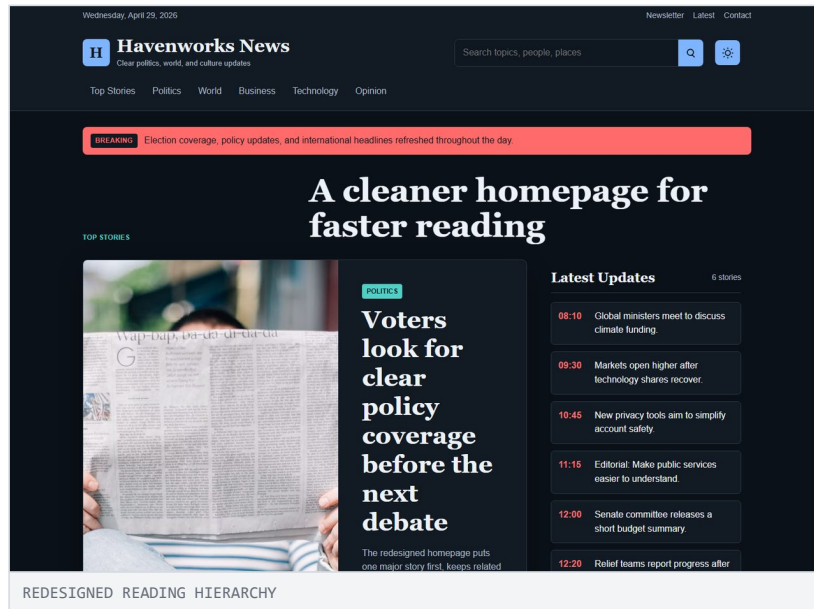


DELIVERED HAVENWORKS

Havenworks Redesign

A structural redesign of a dense news and content site — rebuilding an overloaded page into a clear reading hierarchy without losing the breadth of content the original carried.

Information architecture • HTML / CSS



REGISTER D

Media & video production

Alongside the engineering: commercial video editing sold through Fiverr and Upwork, client campaign work, and the product films that accompany the systems above.

CLIENT WORK

CAMPAIGNS

Client Campaign Editing

Multi-cut campaign delivery for international clients — regional edits, music and no-music variants, vertical reels and HD masters, turned around against short deadlines.

Adobe Premiere Pro · After Effects · CapCut · Colour & sound

PRODUCT FILM

SYSTEMS

Product Walkthrough Films

Screen-recorded walkthroughs for the restaurant, inventory, POS and laboratory systems — used as client demos and onboarding material for staff who will use the software daily.

Screen capture · Motion graphics · Voice-over edit

DOCUMENTARY

LONG FORM

Long-Form & Documentary Edits

Narrative business documentary work and long-form storytelling pieces, cut for pacing and argument rather than montage.

Narrative editing · Archive footage · Sound design

REGISTER E

Foundations & the learning lab

A working directory of deliberate practice — the place where a stack gets learned properly before it goes anywhere near a client.

FULL STACK

CHATBOT

Embeddable RAG Chatbot Platform

The most complete piece in the lab: a containerised support-chat product with a FastAPI backend, Alembic migrations, a React admin dashboard, and a standalone embeddable widget built as its own bundle — the full shape of a SaaS product in miniature.

FastAPI · PostgreSQL · Alembic · Docker Compose · Vite · nginx

COMPUTER VISION

POKI

Body-Pose Game Controller

A Chrome extension that turns webcam body tracking into keyboard input, so browser games are played by moving rather than typing.

Chrome extension MV3 · Pose detection · JavaScript

VISUALISATION

CODING MATHS

Mathematics in Motion

Canvas experiments that render mathematical behaviour as animation rather than notation — built to develop intuition for the geometry underneath graphics and simulation code.

[Canvas API](#) · [JavaScript](#) · [Numerical methods](#)

PRACTICE

[AI](#) / [TensorFlow](#)

Embeddings, Vector Search & TensorFlow

Ground-level work on the machinery behind the AI platforms above: building embeddings by hand, implementing vector similarity search, LangChain pipelines, local Llama inference, and a TensorFlow environment for model training.

[TensorFlow](#) · [LangChain](#) · [Embeddings](#) · [Vector search](#) · [Ollama](#)

PRACTICE

[MERN](#) / [Fast](#)

MERN & FastAPI Fundamentals

Deliberate practice across the two stacks that carry most of the production work — React with Vite on the front, FastAPI services on the back, built from an empty folder each time rather than from a generator.

[React](#) · [Vite](#) · [FastAPI](#) · [Node](#)

PRACTICE

[C](#) / [CPP](#) / [CYPHER](#)

Systems & Cryptography Basics

C and C++ from first principles, plus a cipher workshop implementing classical encryption schemes with a browser front-end — the low-level counterweight to a mostly high-level stack.

[C](#) · [C++](#) · [Python](#) · [Classical ciphers](#)

REGISTER F

Record

Experience, education and capability behind the systems above.

EXPERIENCE

CURRENT

Forward Deployed Engineer & Developer

Regional Blood Centre (RBC) · Mirpur, Azad Kashmir

- Building and deploying the blood bank management system on site, working directly with collection, serology, processing and distribution staff.
- Concurrently developing the laboratory system for the genetics labs of Quaid-e-Azam University.

OCT 2025 – JUL 2026

Operational Manager & Project Manager

FutureSpace, A Software House · Mirpur, Azad Kashmir

- Managed multiple software projects from inception to completion, owning timelines, budgets and quality assurance.
- Led cross-functional teams and introduced agile practices to the delivery process.

SEP – DEC 2025

Technical Associate

Network & Telecom Centre, MUST University

- Backend and full-stack development in Python with a strong focus on Django; also worked within ASP.NET MVC architecture.
- Database design and management across MS SQL and MySQL.
- Collaborated with cross-functional teams to keep network services running without interruption.

2024 – PRESENT

Freelance Software Developer

Independent · Fiverr, Upwork & direct clients

- Designed and built a neurological lab management system in Django for Prof. Sundas Farooq.
- Delivered a restaurant management system and an inventory / point-of-sale platform.
- Contributed to the DHQ Hospital Management System in Mirpur, Azad Kashmir.
- Built websites and brand systems for multiple clients, and sold video editing projects internationally.

EDUCATION

EXPECTED 2030

BSc Computer Science

Virtual University · Mirpur, Azad Kashmir

- In progress alongside full-time development work.

COMPLETING OCT 2026

Google Data Science & AI Program

Coursera

- Applied data science and machine learning foundations.

CAPABILITY

LANGUAGES Python C# JavaScript & TypeScript Kotlin SQL C / C++ HTML & CSS	FRAMEWORKS Django & Django REST FastAPI · Flask ASP.NET Core MVC · EF Core Next.js · React Jetpack Compose Tailwind CSS	DATA & INFRASTRUCTURE PostgreSQL & pgvector MS SQL Server · MySQL Redis · Celery Docker & Compose Alembic migrations NumPy · Pandas · Matplotlib
AI ENGINEERING RAG pipelines & vector search Self-hosted Ollama deployment Multi-agent orchestration Speech-to-text & translation Claude Code · OpenAI Codex TensorFlow · LangChain	DESIGN & MEDIA Brand identity systems Adobe Illustrator & Photoshop Premiere Pro · After Effects CapCut Interface & interaction design	WORKING PRACTICE Forward deployed delivery Project & operations management Agile team leadership Requirements from the field Client communication

Let's put something into production.

Available for software development roles, forward deployed engagements, and freelance system builds. Based in Azad Kashmir, working with clients internationally.

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REFERENCES	Available on request